PDF2Data AI – Complete Project Documentation

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1. Executive Summary

Project Purpose

PDF2Data AI is an enterprise-grade document processing platform that transforms unstructured PDFs into structured, application-ready data using artificial intelligence. This documentation serves as a comprehensive guide for interns joining the development team, covering everything from system architecture to implementation details.

Key Stakeholders

- **End Users** – Organizations dealing with high-volume PDF data extraction (education, healthcare, finance, e-commerce).
- **Development Team** – Backend (Java/Spring), Frontend (React), AI/ML engineers.

Project Goals

1. Automate data extraction from various PDF types (text, scanned, image-based).
2. Achieve >85% accuracy in structured data extraction.
3. Support multiple export formats (JSON, CSV, Excel, SQL, MongoDB).
4. Process 1 000+ pages per hour.
5. Provide a user-friendly dashboard for monitoring and management.

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2. Project Overview

2.1 Problem Statement

Current Challenges in PDF Data Processing:

- Manual data entry is time-consuming and error-prone.
- PDFs are designed for human viewing, not machine processing.
- Different PDF layouts require different extraction approaches.
- Scanned documents require OCR processing.
- Data validation and cleaning is repetitive work.

Cost Impact:

- Average time to manually process 100 PDFs: 40–50 hours.
- Error rate in manual data entry: 5–10%.

2.2 Solution Overview

PDF2Data AI addresses these challenges through:

1. **Automated Document Understanding** – AI detects document structure intelligently.
2. **Smart Data Extraction** – Extracts meaningful fields rather than raw text.
3. **Built-in Validation** – Ensures data quality and accuracy.
4. **Multiple Export Options** – Flexibility in output formats.
5. **Scalable Processing** – Handles large volumes efficiently.

2.3 Key Features

Feature	Description	Benefit
Smart Table Detection	Detects tables without borders	Handles complex document layouts
OCR Integration	Extracts text from scanned images	Processes all PDF types
Multi-page Record Detection	Identifies record boundaries	Accurate bulk processing
Auto-schema Generation	Creates database schemas automatically	Reduces manual setup time
Field Mapping	Recognises equivalent field names	Improves extraction accuracy

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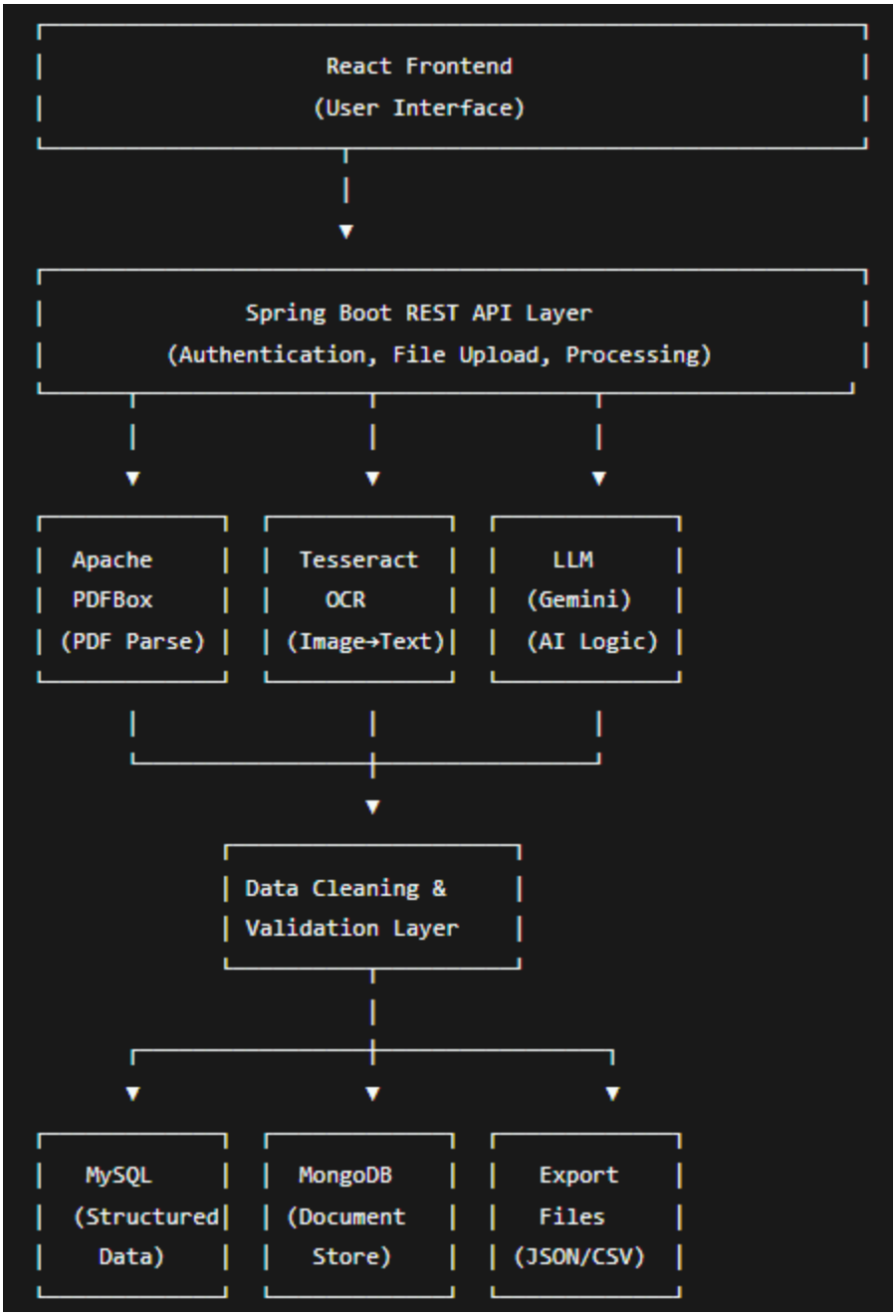
3. System Architecture

3.1 High-Level Architecture



3.2 Component Details

3.2.1 Frontend Layer

- **Technology**: React.js with React Router.
- **Responsibilities**:
  - File upload interface (drag-and-drop).
  - Real-time processing status updates.
  - Data preview and validation interface.
  - Export options selection.

    - Admin dashboard for monitoring.
- **Key Features**:
    - Responsive design.
    - Bulk upload support.
    - Progress tracking with visual indicators.
    - Error display with actionable feedback.

3.2.2 API Layer (Backend)

- **Technology**: Spring Boot 21 with Spring Security.
- **Key Controllers**:
    - `FileUploadController` – handles PDF uploads.
    - `ProcessingController` – manages document processing.
    - `ExportController` – handles data export.
    - `AdminController` – system monitoring endpoints.
- **Security**: JWT-based authentication with role-based access control.

3.2.3 AI Processing Pipeline

text

Input PDF → Layout Analysis → Text Extraction → AI Structure Detection → Data Validation → Export

1. Layout Analysis – Identifies document structure (tables, forms, paragraphs); detects page boundaries and sections; handles both text and image-based PDFs.

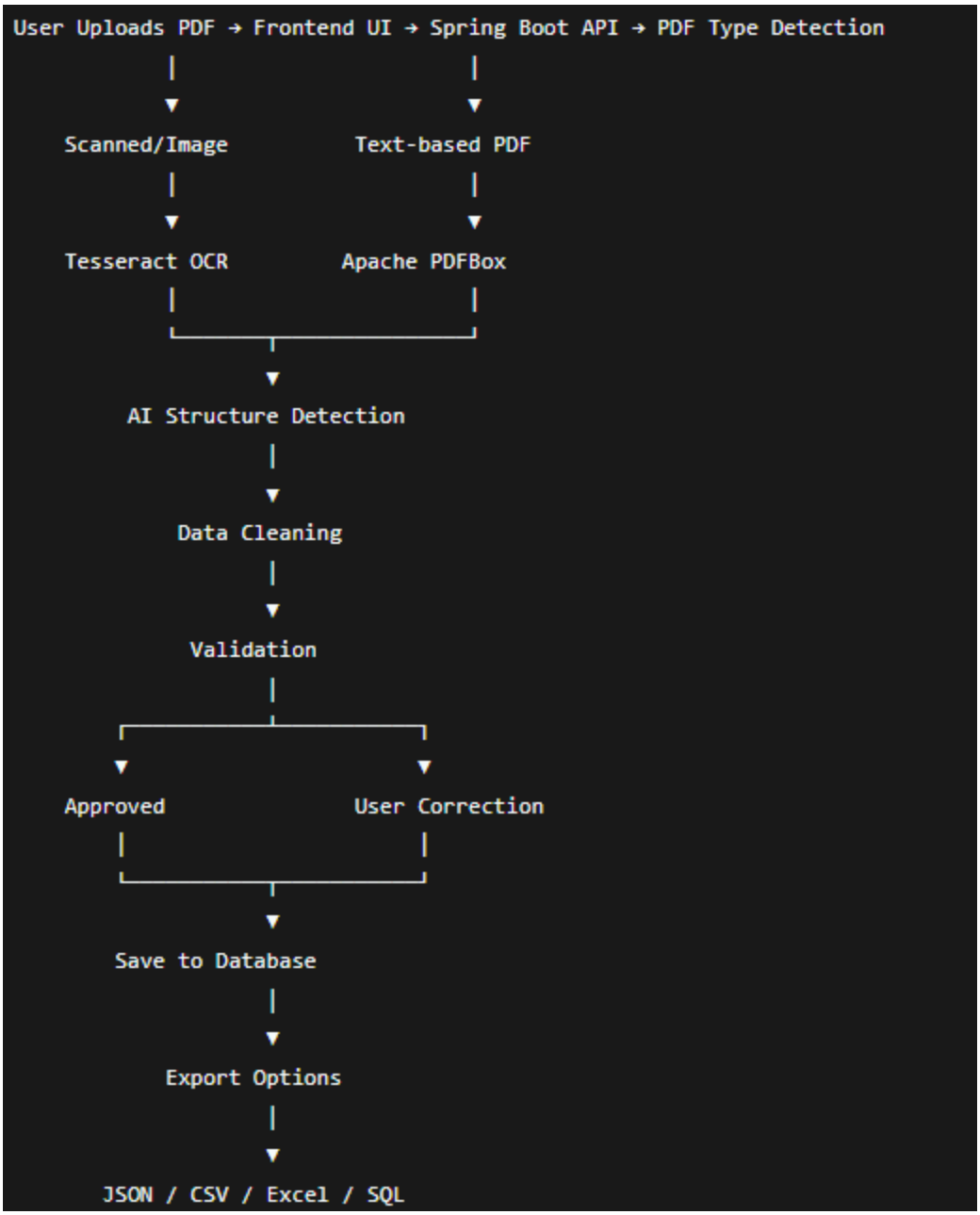
2. Text Extraction – Uses Apache PDFBox for text-based PDFs; uses Tesseract OCR for scanned/image-based PDFs; preserves formatting and layout information.

3. AI Structure Detection – Uses Gemini/OpenAI to understand document context; identifies key-value pairs, tables, and lists; maps extracted data to standardized fields.

4. Data Cleaning & Validation – Removes noise (extra spaces, special characters); standardises date/time formats; validates emails, phone numbers, and other fields; handles missing or incomplete data.

5. Export & Storage – Multiple export formats (JSON, CSV, Excel, SQL); direct database insertion (MySQL, MongoDB); supports both on-demand and scheduled exports.

3.3 Data Flow Diagram



4. Technology Stack

4.1 Backend Technologies

Technology	Purpose	Key Notes
Java	Core programming language	Use LTS version
Spring Boot	Application framework	Use Spring Initializr
Spring Security	Authentication & Authorization	JWT-based
Spring Data JPA	Database ORM	Hibernate behind scenes
Spring AI	LLM integration	OpenAI/Gemini support
Apache PDFBox	PDF processing	Handles text extraction

Technology	Purpose	Key Notes
Tesseract OCR	Image text extraction	Use with Tess4J wrapper
MySQL	Relational database	For structured data
MongoDB	Document database	Optional for flexible data
RabbitMQ	Message queue	For async processing
Redis	Caching	Session management

4.2 Frontend Technologies

Technology	Version	Purpose
React	18.x	UI framework
Redux/Context API	Latest	State management
React Router	6.x	Navigation
Axios	Latest	HTTP requests
React Dropzone	Latest	File upload
Tailwind CSS	Latest	Styling
Chart.js	Latest	Analytics dashboard

4.3 Development Tools

Tool	Purpose
IntelliJ IDEA	Java development
VS Code	Frontend development
Docker	Containerisation
Kubernetes	Orchestration
Postman	API testing
Git	Version control
JUnit 5	Testing framework
Mockito	Mock testing

5. Common Pitfalls & Solutions

Issue	Root Cause	Solution
Missing text in PDF	PDF contains images/scans	Enable OCR processing
Garbled text	Wrong encoding	Auto-detect encoding or use UTF-8
Tables not detected	Missing borders or alignment	Enhance table detection algorithm
Slow processing	Large files or complex layouts	Use async processing with queue
Memory errors	Processing many large files	Implement pagination and streaming
Wrong field mapping	Similar field names	Add validation rules and synonyms
Missing fields	Incomplete prompt	Improve prompt engineering
Inconsistent formatting	Different document styles	Train on multiple document types
Hallucinations	LLM uncertainty	Add confidence scores and validation